
Learning Goal-Conditioned Robotic Manipulation under Sparse Rewards with Hindsight Experience Replay

Elliot Markovich and Daria Goptsii

McGill University
Montreal, Canada

Abstract

Learning goal-conditioned policies under sparse rewards is challenging due to difficult exploration. We study this problem in the FetchReach environment, evaluating off-policy methods (DDPG, TD3, SAC). While all methods succeed with dense rewards, only SAC learns under sparse rewards, and with poor sample efficiency.

We address this using Hindsight Experience Replay (HER), which relabels goals to generate additional training signal. HER enables successful learning and substantially improves sample efficiency, while prioritized HER further accelerates convergence. In contrast, curriculum learning and goal sampling provide limited or negative gains.

Overall, our results show that goal relabeling is sufficient to solve simple sparse-reward tasks, with more complex methods offering diminishing returns.

1 Introduction and Literature Overview

Reinforcement learning (RL) performs well in simulated control tasks but remains challenging in robotic settings with sparse rewards, where agents receive little feedback unless they succeed. This leads to poor sample efficiency and unstable learning.

Goal-conditioned reinforcement learning addresses this by learning policies $\pi(a \mid s, g)$ that generalize across goals, as in Universal Value Function Approximators (UVFAs) Schaul et al. [2015]. We study this setting in the FetchReach environment Plappert et al. [2018], Farama Foundation [2024], where a robotic arm must move its end-effector to a target position in 3D space.

Sparse rewards provide only a binary success signal, making exploration difficult, while dense rewards provide smoother optimization signals Andrychowicz et al. [2017]. We evaluate standard off-policy methods (DDPG, TD3, SAC) and show that they fail under sparse rewards despite succeeding in the dense setting.

We then incorporate Hindsight Experience Replay (HER) Andrychowicz et al. [2017], which relabels trajectories with achieved goals to generate additional training signal. We further evaluate prioritized HER and curriculum-based methods to assess whether more complex strategies improve performance beyond goal relabeling.

2 Environment

We evaluate our methods on the FetchReach environment Plappert et al. [2018], Farama Foundation [2024], where a 7-DoF robotic arm must move its end-effector to a target position in 3D space.

The environment follows a goal-conditioned formulation Schaul et al. [2015]. At each timestep, the agent receives the observation s , achieved goal g_{achieved} (current end-effector position), and desired goal g_{desired} (target position).

Rewards are defined based on the distance between achieved and desired goals Farama Foundation [2024]. We consider:

- **Sparse:** 0 on success, -1 otherwise.
- **Dense:** negative distance to the goal.

This formulation enables techniques such as HER Andrychowicz et al. [2017].

3 Baseline Comparison

3.1 Experimental Setup

We compare DDPG Lillicrap et al. [2015], TD3 Fujimoto et al. [2018], and SAC Haarnoja et al. [2018] in a goal-conditioned setting Schaul et al. [2015]. All methods share the same architecture and replay buffer, with inputs formed by concatenating observations and goals.

Performance is evaluated using success rate and sample efficiency, measured as the number of episodes required to reach 90% success. Each method is evaluated under both dense and sparse rewards on FetchReach.

3.2 Dense Rewards

Table 1 summarizes performance under dense rewards. All methods achieve near-perfect final success, but differ in sample efficiency.

Table 1: Performance on FetchReach with dense rewards.

Method	Final Success	Last-5 Mean	Episodes to 90%
DDPG	1.00	1.00	180
TD3	1.00	0.98	220
SAC	1.00	1.00	90

All methods solve the task, indicating optimization is not limiting. However, SAC is more sample-efficient, reaching 90% success in 90 episodes versus 180 (DDPG) and 220 (TD3), motivating its use as the primary baseline.

3.3 Sparse Rewards

Table 2 shows performance under sparse rewards.

Table 2: Performance on FetchReach with sparse rewards (no HER).

Method	Final Success	Best Success	Episodes to 90%
DDPG	0.00	0.20	—
TD3	0.10	0.10	—
SAC	1.00	1.00	350

In contrast to the dense setting, DDPG and TD3 fail to learn, never exceeding 20% success. SAC is the only method that succeeds, but requires 350 episodes to reach 90% success, nearly $4\times$ slower than in the dense setting.

These results indicate that exploration, rather than function approximation, is the primary bottleneck. We therefore introduce Hindsight Experience Replay (HER) to improve learning in the sparse setting.

Additional training curves, including evaluation success, return, and goal distance for all baseline methods under both dense and sparse rewards, are provided in Appendix A.

4 Goal Relabeling and Curriculum Methods

4.1 SAC with HER

We augment SAC with Hindsight Experience Replay (HER) Andrychowicz et al. [2017]. Table 3 compares SAC with and without HER.

Table 3: Effect of HER on SAC (sparse rewards).

Method	Final Success	Episodes to 90%
SAC	1.00	350
SAC + HER	1.00	190

HER reduces the number of episodes required to reach 90% success from 350 to 190, a $1.8\times$ improvement in sample efficiency. This demonstrates that relabeling unsuccessful trajectories provides a strong learning signal in sparse-reward settings.

4.2 Prioritized HER

We further evaluate prioritized HER, which biases replay toward more informative transitions.

Table 4: Comparison of HER variants (sparse rewards).

Method	Final Success	Episodes to 90%
SAC + HER	1.00	190
SAC + Prioritized HER	1.00	150

Prioritized HER further improves sample efficiency, reaching 90% success in 150 episodes. This corresponds to a $\sim 21\%$ improvement over standard HER, indicating that selecting more informative relabeled goals accelerates learning.

4.3 Curriculum and Goal Sampling

We evaluate curriculum learning and GoalGAN-style goal sampling methods, which aim to improve exploration through intermediate goals.

Table 5: Curriculum and goal sampling methods (sparse rewards).

Method	Final Success	Episodes to 90%
SAC + HER	1.00	190
+ Linear Curriculum	1.00	180
+ Adaptive Curriculum	1.00	200
+ GoalGAN	1.00	210

These methods do not outperform prioritized HER. Linear curriculum slightly improves convergence (180 episodes), while adaptive curriculum and GoalGAN degrade performance. Overall, HER provides sufficient exploration, and additional mechanisms add complexity without benefit.

Full training curves for HER and its variants, including evaluation success, return, and goal distance, are provided in Appendix A.

5 Discussion

Our results indicate that the main challenge in FetchReach is the lack of informative reward signals rather than function approximation.

Among HER-based methods, prioritized HER further improves sample efficiency by biasing training toward more informative relabeled transitions. This suggests that, beyond simply increasing the quantity

of training data, the quality and relevance of replayed experiences play an important role in learning performance.

In contrast, curriculum learning and goal sampling methods provide limited benefit. Since FetchReach is a relatively simple task with a low-dimensional goal space, HER already generates sufficiently informative training signals. Additional mechanisms for generating intermediate goals introduce extra complexity without improving final performance, and in some cases reduce sample efficiency. This indicates that curriculum methods may be more beneficial in higher-dimensional or long-horizon tasks, where exploration remains challenging even with HER.

Overall, our findings suggest that HER is the dominant factor in enabling learning in sparse-reward goal-conditioned tasks, while more complex extensions provide diminishing returns in simple environments. This in turn suggests that in low-dimensional goal spaces, relabeling provides sufficient coverage of the goal distribution, reducing the need for explicit exploration strategies.

6 Conclusion

We studied goal-conditioned learning under sparse rewards in FetchReach and showed that standard off-policy methods fail in this setting. HER enables successful learning by relabeling goals, and prioritized HER further improves sample efficiency.

More complex methods such as curriculum learning and goal sampling do not provide additional benefits, indicating that improving data efficiency through goal relabeling is more important than increasing algorithmic complexity in simple sparse-reward tasks.

7 Future Work

While our experiments provide clear insights into sparse-reward learning in FetchReach, several directions remain for future investigation.

First, our study is limited to a single task. Extending the evaluation to more complex environments such as FetchPush, FetchSlide, and FetchPickAndPlace would provide a more comprehensive understanding of how different methods scale with task difficulty. These tasks introduce additional challenges, including object interaction, longer horizons, and more complex dynamics, which may increase the importance of exploration strategies beyond HER. In particular, curriculum learning and goal sampling methods may become more effective in these settings, where exploration remains difficult even with goal relabeling.

Second, hierarchical reinforcement learning represents a promising direction for improving performance in more complex tasks. Methods such as HIRO Nachum et al. [2018] introduce temporal abstraction by learning high-level policies that propose subgoals for a lower-level controller. While such approaches were not explored in this work, they may be better suited for tasks requiring multi-stage planning and long-horizon coordination, such as pick-and-place manipulation.

8 Limitations

This study has several limitations that may affect the generality of the results.

First, experiments were conducted under a limited compute budget, restricting training time and the extent of hyperparameter tuning. As a result, some methods may not have been fully optimized. Second, results are averaged over a small number of random seeds, which limits the reliability of variance estimates. Third, hyperparameters were not exhaustively tuned for all methods, particularly for more complex approaches such as curriculum learning and goal sampling, where performance may be sensitive to design choices.

Despite these limitations, our results provide a consistent comparison of methods and highlight the dominant role of goal relabeling in sparse-reward learning.

9 Contributions

Both group members collaborated on all aspects of the project. However, Daria’s work generally focused more on coding whereas Elliot’s work focused more on report writing and creating the presentation.

A Additional Results

We provide additional training curves for all methods. Each plot shows performance over training episodes.

A.1 Baseline Methods

Figures 1, 2, and 3 show results for DDPG, TD3, and SAC under both dense and sparse rewards.

A.2 HER and Variants

Figures 4, 5, and 6 show results for SAC with HER and its variants (prioritized HER, curriculum learning, and GoalGAN) under sparse rewards.

References

- Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. In *Advances in Neural Information Processing Systems*, 2017.
- Farama Foundation. Gymnasium-robotics: Fetch environments documentation. <https://robotics.farama.org/envs/fetch/>, 2024. Accessed: 2026-04-24.
- Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1587–1596, 2018.
- Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning*, pages 1861–1870, 2018.
- Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015.
- Ofir Nachum, Shixiang Gu, Honglak Lee, and Sergey Levine. Data-efficient hierarchical reinforcement learning. In *Advances in Neural Information Processing Systems*, 2018.
- Matthias Plappert, Marcin Andrychowicz, Alex Ray, Bob McGrew, Bowen Baker, Glenn Powell, Jonas Schneider, Josh Tobin, Maciek Chociej, Peter Welinder, et al. Multi-goal reinforcement learning: Challenging robotics environments and request for research. *arXiv preprint arXiv:1802.09464*, 2018.
- Tom Schaul, Daniel Horgan, Karol Gregor, and David Silver. Universal value function approximators. In *Proceedings of the 32nd International Conference on Machine Learning*, pages 1312–1320, 2015.

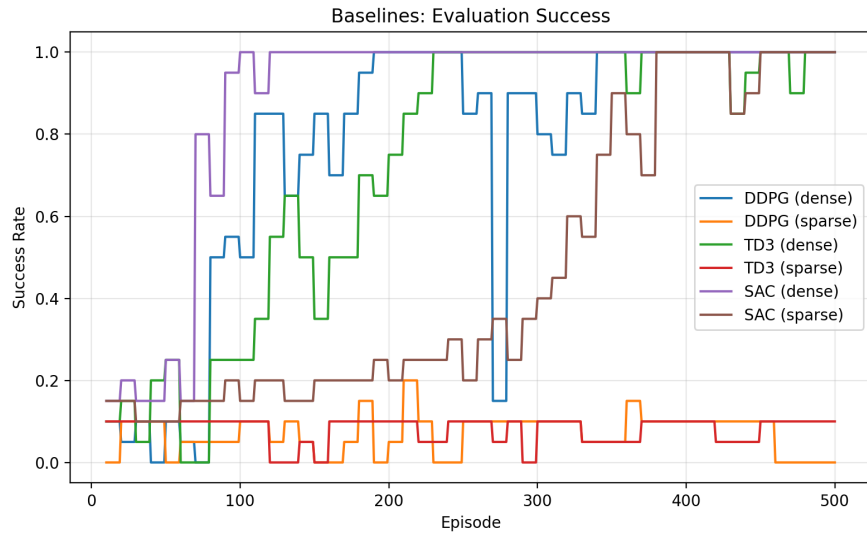


Figure 1: Evaluation success rate for baseline methods (dense and sparse rewards).

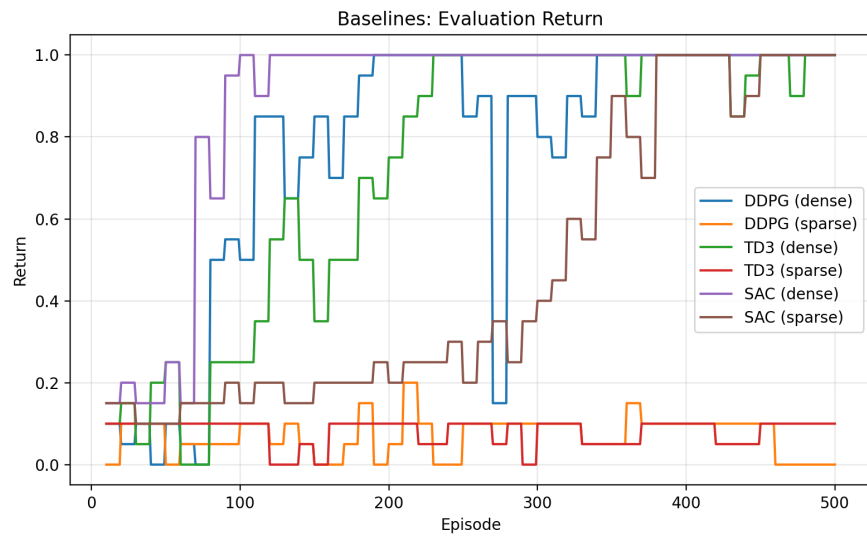


Figure 2: Evaluation return for baseline methods (dense and sparse rewards).

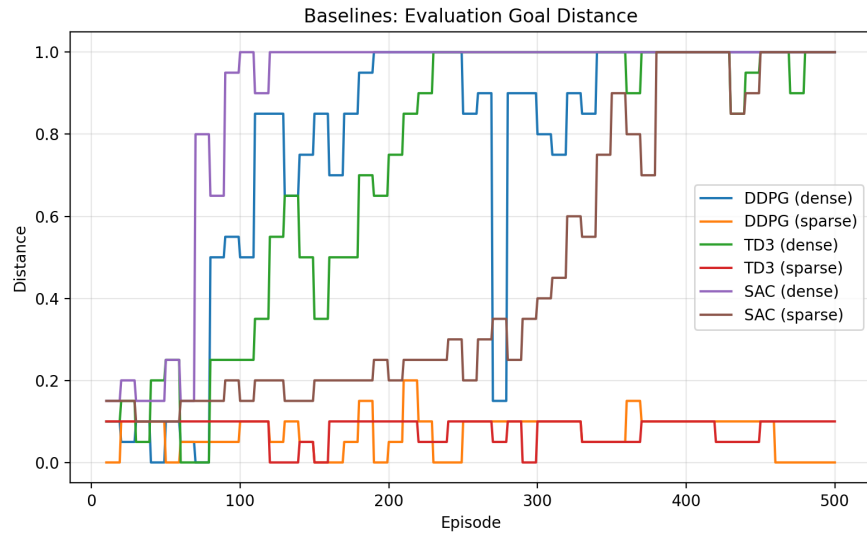


Figure 3: Evaluation goal distance for baseline methods (dense and sparse rewards).

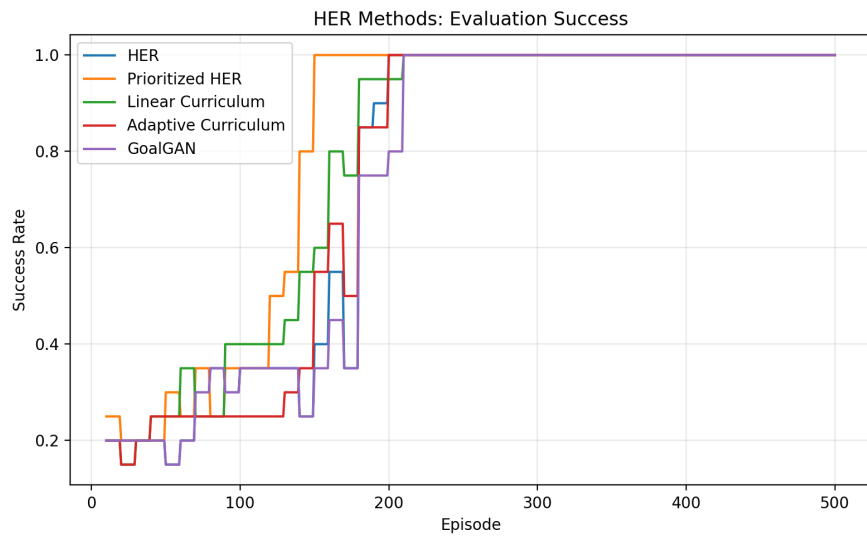


Figure 4: Evaluation success rate for HER-based methods (sparse rewards).

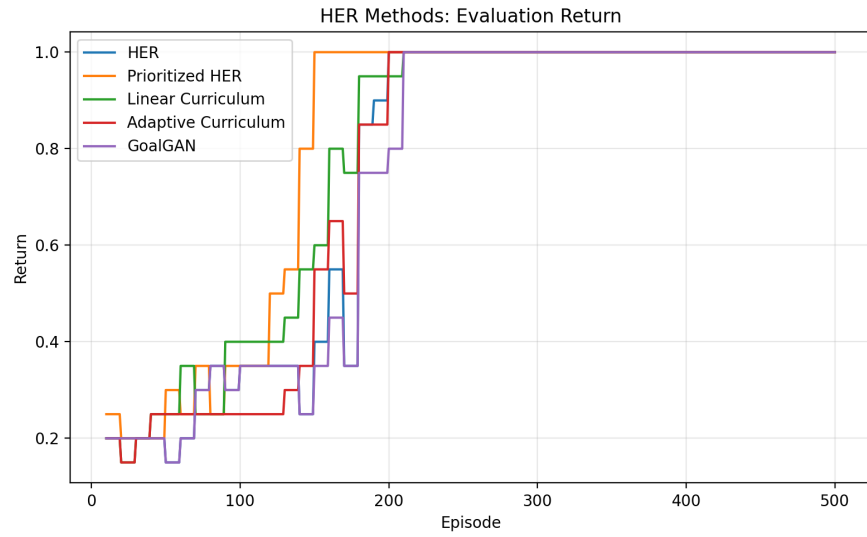


Figure 5: Evaluation return for HER-based methods (sparse rewards).

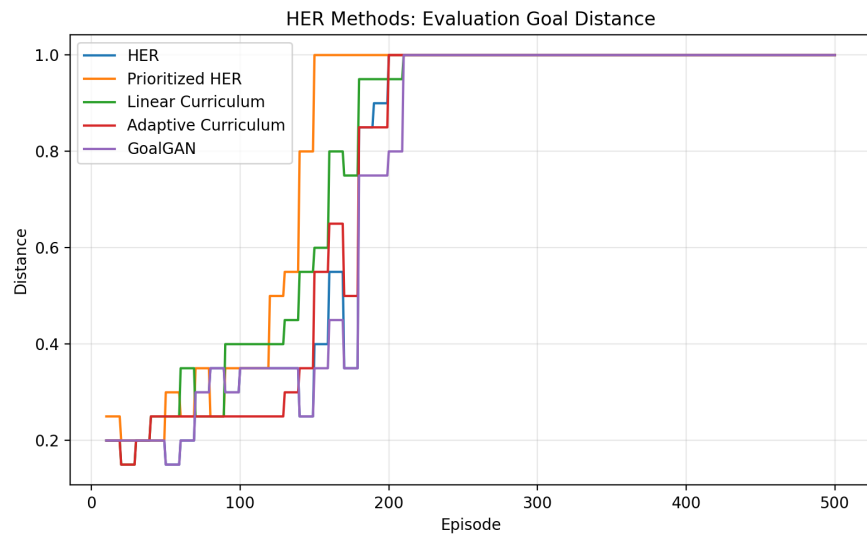


Figure 6: Evaluation goal distance for HER-based methods (sparse rewards).